RYAN J. ERICKSON

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EDUCATION

University of Colorado, Boulder *August 2021 - (expected) October 2025*
Geography, MA

University of California, Santa Barbara *September 2018 - May 2020*
Geography - GIS Emphasis, BA

University of California, Santa Barbara *September 2017 - May 2020*
Physics

Chaffey Community College *August 2015- June 2017*
International General Educational Transfer Curriculum (IGETC), AA

RESEARCH INTERESTS

Geographic Informational Science (GIS) *January 2019 - current*
Spatial-temporal analysis, mapping software, data science

Remote Sensing of the Environment *January 2019 - current*
Satellite data acquisition, Big Data queries, lattice structures

Quantitative Physical Geography *January 2019 - current*
Atmospheric Chemistry, Meteorology, Hydrology. Geomorphology

Computer Science and Technology *August 2021 - current*
Programming and engineering of computational systems.

Qualitative Human Geography *March 2022 - current*
Public health, exposure modeling, DAGs, survey composition and analysis

Physics *September 2018 - May 2020*
Thermodynamics, Electrodynamics, Quantum Mechanics, Astrophysics, Biophysics

RESEARCH EXPERIENCE

STAR Lab Masters Student *August 2021 - October 2025*
Spatiotemporal Pattern Analysis and Research Laboratory - Guofeng Cao

- Composed 110+ page thesis detailing machine learning methods for surface ozone modeling in urban areas.
- Combined EPA monitor data, satellite imagery, and census data for potential exposure assignment program.
- Specialized in spatiotemporal patterns and dynamics for numerous natural, social and technical systems.

NC CASC Graduate Research Assistant *May 2025 - August 2025*
North Central Climate and Science Center - Heather Yocum, Matthew Rossi, and Alison Post

- Composed blog for NC CASC website summarizing our work and future directions.

- Provided quick displays in Google Earth Engine to convey cutting edge research options.
- Attended meetings to summarize research methods, provide advice, and professional insights into graduate student life for pending applicants.

NHC Graduate Research Assistant *January 2025 - May 2025*
Natural Hazards Center - Lori Peek, Mary Painter and Kate McNeely

- Gathered more than 2,000 documents for further analysis, sorting, and inclusion into future literature by Lori Peek (Third Assessment of Natural Hazards).
- Used SQL to search 20+ disciplines on EBSCOHost, Web of Science, CU Scholar, and Google Scholar
- Developed a python script to access the above APIs (requests, rispy, pandas, numpy)

Earth Lab Graduate Research Assistant *August 2024 - December 2024*
Partnered with the CIRES - Cibele Amaral and Adam Mahood

- Debugged and assessed FiredPy, a Python CLI program for classifying fire events with the MODIS Burned Area Product (Collection 6).
- Engineered new spatial strategies for solving spatial-temporal edge case scenarios.
- Optimized 1TB+ of county-level data for acquisition and display.

IBS Graduate Research Assistant *January 2022 - December 2024*
Institute of Behavioral Science - Stefan Leyk and Colleen Reid

- Collaborated with external university professionals, translated their advice to peers and research personal.
- Applied spatial-temporal methods to build an activity space of survey participants.
- Assigned modeled air pollution exposures to individuals exposed to the Marshall Fire after December, 2021 in Superior, CO.

MOVE Lab Research Assistant *March 2020 - August 2020*
MOVE Research Group - Somayeh Dodge

- Collaborated with external university professionals, translated their advice to peers and research personal.
- Applied spatial-temporal methods to build an activity space of survey participants.
- Assigned modeled air pollution exposures to individuals exposed to the Marshall Fire after December, 2021 in Superior, CO.

TEACHING EXPERIENCE

GIS: Mapping *January 2024 - May 2024*
Instructor: Sarah Kelly

- Topics included how to build a spatial database, best practices for processing environmental/social data, and basic visual analytics.
- Responsible for managing labs and discussions for for 150+ students across 4 lectures.
- Proctored midterm, final exam, and bi-weekly quizzes for the instructor.

Statistics and Geographic Data *January 2023 - May 2023*
Instructors: Guofeng Cao, Morteza Karimzadeh *August 2021 - December 2021*

- Conveyed basic statistical concepts, methods and tools commonly used to detect, characterize and explain geographic data.
- Taught, advised, and graded undergraduate work for 80+ students across 2 labs.

- Moderated, amended, and reorganized class material for students with various instructor personalities.

GIS: Mapping (Summer) *May 2022 - June 2022*
Instructor: Sarah Kelly *May 2023 - June 2023*

- Offered additional summer instructional hours for 250+ students across 4 lectures.
- Summarized and truncated full labs into smaller assignments with expected less completion time.
- Incorporated material more pertinent to real-world events (voter analysis, SARS COVID-19, etc.).

Statistics and Geographic Data *January 2022 - May 2022*
Graduate Part Time Instructor (GPTI): Kylen Solvik

- Provided methods, course material, and summaries of prior instructor methods to GPTI.
- Organized office hours, classroom times, and lab sections to provide 12/7 assistance.
- Gave lectures as the official instructor for over 120 students across 2 classes.

PUBLICATIONS

Erickson, R., Cao, G., Isaacs, R., & Leyk, S. (2025, October). *Integrating Machine Learning and Geostatistics for High-Resolution Surface Ozone Mapping: A Case Study in Arizona* [Master's thesis, University of Colorado (in review)].

SPECIALIZATIONS

Programming and Code	Python, R, Javascript, SQL, Bash, MATLAB, XML/GML
Python Packages	Matplotlib, Miniconda, Sci-kit Learn, geemap, etc.
Software & Tools	Docker, Github, ArcGIS (ESRI Suite), QGIS, ENVI
	Microsoft Office, HTML, LaTeX, Google Earth Engine

GRANTS & FELLOWSHIPS

STAR Lab Masters Graduate Student	*August, 2021 - October 2025*
CIRES Early Career Assembly Member	*May, 2022 - May, 2026*
Climate Adaptation Scientists of Tomorrow Fellow	*May, 2025 - August, 2025*
CU Population Center (CUPC) Research Fellow	*August, 2022 - December 2022*

CERTIFICATIONS

FAA CFR Part-107 Remote Pilot	*May 2024 - (expires May 2026) current*
SCUBA Certification	*August 2017 - current*
Red Cross EMS First Responder	*January 2016 - (expired) January 2024*

AWARDS AND HONORS

Chaffey Community College Academic All-American	*May 2017*
National Honor Society	*January 2015*